Fiscal Year:	FY 2008	**Task Last Updated:**	FY 05/02/2008
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Project Title:	Differential Gene Expression in Endothelial Cells Cultured in Rotating Wall Vessel Bioreactors & Exposed to Low Laminar Shear Stress		
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Comments:			
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Flight Assignment:	NOTE: Changed Division and Discipline/Program to FSB as of FY2006, per program changes at that time, per J. Cohen/HQ and JSC/J. Dardano and previous information fr D. Risin/JSC in Mar 2006 (jvp 6/2009) NOTE: Funding reinstated to 11/30/2007, per J. Cohen (HQ, 8/2006)		
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Task Description:	Rotating Wall Vessel Bioreactors offer a unique biotechnology approach to 3-D cell culture and advanced tissue engineering applications. In these fluid-filled Bioreactors, particles suspended in the culture medium rotate with the fluid in a quasi-solid body configuration. Cells and cell assemblies in this venue are in continual free fall and thus concomitantly exposed to low laminar shear stress (~ 0.5 dynes/cm2) and also to "simulated or modeled microgravity ", i.e. to a time-averaged, randomized gravitational force (~0.01 x g). To date, the contribution of each of these factors, especially of reduced laminar shear stress, has not been thoroughly assessed experimentally. Based on reports in the literature and our own preliminary data, we hypothesize that the component which simulates/models microgravity by "randomizing" the gravitational vector, rather than low, laminar flow–induced shear stress, may be instrumental in modulating gene expression and functional differentiation of cells/cell assemblies in RWV Bioreactors. In testing our hypothesis, we propose to use genomics, proteomics and bioinformatics approaches for analyzing patterns of differential expression of genes and gene products in human endothelial cells cultured for various amounts of time (4 h –20 days) in RWV Bioreactors, with controls maintained static monolayers, static suspension cultures, or as monolayers in a parallel plate flow chamber exposed to fluid shear stress of 0.5 dynes/cm2. In addition, by using the Perfused Rotating Wall Vessel Reactors, we will be able induce controlled levels of fluid shear stress levels in these Bioreactors. Endothelial cells are chosen for this study, since a) the fluid-shear dependent gene activation in these cells is well established and characterized b) altered fluid dynamics and mechanical forces, such as present in RWV Bioreactors (and space flight?) are known risk factors in the etiology of diminished cardiac and cardiovascular function, specifically in the etiology of atherosclerosis. The objectives of this study are as follows: Specific aim 1: Compare differential gene expression in human microvascular endothelial cells cultured respectively, as monolayers under static conditions, or exposed to low levels of steady shear stress, and in suspension culture in Rotating Wall Vessel Bioreactors. Specific Aim 2: Compare differential protein expression in human microvascular endothelial cells cultured under the same conditions described above in Specific Aim1. Specific aim 3: Use modern bioinformatics approaches for identifying functional correlations between expression patterns of genes and gene products, specifically of signaling molecules and pathways, which uniquely are expressed (only) in RWV Bioreactors Specific Aim 4: Use established molecular biological approaches for identifying a unique RWV –sensitive response element in the promoter region of one or more RWV-responsive genes. While being responsive to several different stipulations of the soliciting NRA (01-OBPR-08), the objectives of this ground-based proposal is of relevance to NASA's mission, as outlined in the Critical Path Roadmap, specifically, Risk 6 Diminished Cardiac and Vascular Function. In this context our study adresses Enbling Questions 6a-6c: Does long-duration space flight lead to diminished cardiac function? What mechanisms are involved? Is the process reversible? • In studying the workings of NASA developed RWV Bioreactor Biotechnology we are focusing on a) elucidating the value of low shear culturing and b) studying differential endothelial gene expression in RWV Bioreactors. • This study will distinguish between the effects of low shear and that of other parameters in the workings of RWV Bioreactors, specifically those components that may represent a ground-based microgravity analog. Furthermore, the comprehensive approach will lead to the identification of candidate mechanism and genes/gene products that are uniquely responsive to the modeled microgravity component of the RWV environment. • The outcome of this study will facilitate rationally designing meaningful space flight experiments, aimed at deciphering the mechanisms by which cells and organisms sense and transduce microgravity. Thus, this study provides a strong link and a clear path to flight research on the International Space Station and beyond for long-duration space flight. • The proposed studies entail strong educational components, both for our students as well as for students from local/regional schools.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed study will enable the rational use of an important flight hardware, the RWV Bioreactor system, for ground based studies addressing important risks that figure prominently in the Critical Road Map, such as such as cardiac and vascular functions, impaired immune responses, etc. In addition, the strong educational component of this study will be leveraged for outreach activities, to raise the level of awareness and enthusiasm for NASA-related projects amongst high school and college students.
	In the final year of this grant we have continued to make significant progress towards all four specific aims. In order to identify from all genes uniquely identified as RWV related those unique genes, gene products and signaling pathways which are potentially affected by the component of modeled microgravity (MMG), we carried out a series of experiment with identical cell preparations in the different venues and analyzed their differential gene expression using Codelink microarrays (as detailed in the Methods section) followed by stringent statistical analysis of the data. As an example, as a first validation of the approach, we compared differential gene expression between static and flow conditions in the PPFC. The number of statistically significantly altered genes depended very much on the stringency of the analysis. Of the ~20,000 genes sported on the Codelink microarray , only ~ 12,000 genes were found to be reproducibly analyzable in 6 independent experiments. Of those we saw a statistically significant change in 790 genes at conventional stringency (95% confidence); 657 genes were found to be differentially expressed under high stringency (99% confidence, while only 60 genes were found to be differentially expressed at extreme stringency (99.9% confidence). By comparison, analysis of the data by high stringency ANOVA (99% confidence) yielded 552 genes that were differentially expressed between PPFC and static conditions. Based on these results, all our subsequent analyses were based on ANOVA test with a 99% confidence ($p < 0.01$) Using these state-of-the art tools, we have identified a number of genes which the diverse venues have in common, but also others, which are uniquely expressed in each of the venues investigated. Importantly, we have identified a limited number of genes (17) genes which seem to be uniquely are expressed (only) in RWV Bioreactors, and which may be related to the modeled microgravity (MMG) component in this venue. Two genes in this group, AQP3 (Aquaporin 3) and BMP1(Bone morphogenetic protein-1) are of specific interest. As we have shown previously, RWV conditions favor the neuroendocrine differentiation of PC12 pheochromocytoma cells with concomitant downregulation of their neuronal phenotype. The unique expression of these two genes might explain some of our prior findings. AQP3 is a water channel protein that has been shown to play a role in the development of glandular tissue in the rat and is expressed in many tissues, including the basolateral membrane of pancreatic ducts and articular chondrocytes. BMP1 is a calcium dependent metalloprotease not known to be structurally similar to any other human growth factors or the other members of the BMP family. In humans BMP1 has been shown to both induce cartilage formation in vivo and inhibit neurogenesis. The upregulation of either or both of these two factors, if more specifically demonstrated to play roles in developing the RWV phenotype, potentially helps explain the spontaneous differentiation of PC12 cells towards their neuroendocrine (rather than neural) phenotype, and the increased GAG production and layered, organized structures

Task Progress:	seen in chondrocyte cultures. In evaluating possible functional relationships between the differentially expressed genes we used advanced systems biology-based bioinformatics tools, such as Ingenuity and PAINT. The Ingenuity Pathways Analysis application makes use of the Ingenuity Pathways Knowledge Base, the world's largest proprietary and commercial curated database consisting of millions of individually modeled relationships between proteins, genes, complexes, cells, tissues, drugs, and diseases. PAINT (A Promoter Analysis and Interaction Network Generation Tool) is a unique open source data base and tools for Regulatory network identification. Both these bioinformatics tools in combination were to establish potential associations between gene circuits and signaling pathways. Using the differential gene expression between static and low laminar shear stress flow activated cells in the PPFC as our initial paradigm for validation of the system, the analysis by both Ingenuity and PAINT revealed a central role for the transcription factor c-myc as the most common element in modulating differential gene expression). By contrast no such relations could be established for the MMG component in the RWV environment, presumable because of the small number of candidate genes that are uniquely attributable to be affected by MMG. In addition to the genomics analysis we also carried out some proteomics-based studies. As a first step, total cell extracts were separated by 2-D gel electrophoresis and analyzed by differential protein profiling using fluorescent DIGE (differential in gel electrophoresis). Of the total number of 1310 resolved protein spots, 71 (5.4%) were decreased and 36 (2/7%) were increased by flow. In addition to global proteomics analysis, we also investigated at the protein level how the RWV conditions differentially affect the activation of specific signaling pathways. For this, we first analyzed the activation state of phosphorylation dependent protein kinases, using a commercially available proprietary protein kinase profiling tool (KPKS from KINEXUS). There is a similarity in the overall activation state/level of phosphorylation of several secondary messenger/transcription factors, such as JNK, FAK and GSK-3, as well of some of the PKC isoforms under static conditions and in bags, which is significantly different for m their activation profiles in the RWV. To further substantiate our conclusions in the human microvascular endothelial cell system, we have also directly measured the time course of activation of select signaling pathways, beta-testing a commercial phosphor array from BIOSOURCE. The kinetic results confirm our prior findings, that certain phosphorylation dependent cell signaling (e.g. through hormone receptors, such as EGFR or NGFR) is spontaneously upregulated for prolonged periods of time in RWV bioreactors. Specifically, the increase in AKT phosphorylation independently confirms our findings for an increased activity/expression of GSK-3. Taken together, our investigation of cell signaling in the RWV suggests that the state of activation of select signaling pathways/second messenger systems under static conditions and in a bag (aggregation) is quite similar in each of these venues, but qualitatively (and quantitatively different form when the cells are cultured in the HARV-type RWV. Furthermore, these data are in line with our recent findings on the transient impairment of the immune response in human lymphocytes in the RWV environment. According to our data, the individual components of various post-receptor signal transduction cascades are not entirely abolished, but exposure, to the MMG causes quantitative imbalance in the level of activation of individual components. Finally, and beyond the original goals of the proposal, we developed a novel “horizontal microscopic” module, that allows us to analyze in real time particle movement and, especially, cell aggregation in HARV-type RWV Bioreactors. The rational for this approach (part of a new MD/PhD thesis) is to a) develop algorithms which will allow the automated rotational control of the RWV Bioreactors depending on the state of aggregation of the rotating particles ad b) to develop and validate computational algorithms for modeling cell aggregation in 3-D. Finally, as a new technological advance, we are working on a novel optical system to determine in real-time the aggregation kinetics in RWV Bioreactors. We anticipate that the novel methodology and algorithms developed in this work will enhance our ability for non-invasive, remote monitoring and data analysis during the exploration to Moon and Mars.

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